Seat Number | | | | | | |

MEGH-27

MCA-417 : Artificial Intelligence
(1212410007)

Total Page : 1]

Time: 3 Hours **Max. Marks :60**

Instruction to Candidates:

1. Do not write anything on question paper except Seat No.
2. All questions are compulsory.
3. Figures to right indicate full marks.
4. Students should note, no supplement will be provided.
5. Graph or diagram should be drawn with the black ink pen or black HB pencil.

1. Answer **any four** of the following. **12**

i) Explain the scope of Symbolic AI with an example.
ii) What is a Rule-Based System in AI? Provide an example.
iii) What is Tabu Search. Explain with example
iv) Explain the SSS* algorithm
v) Describe the concept of State Space Representation
vi) Explain AI techniques with its type and example

2. Attempt **any three** of the following. **12**

i) Differentiate between DFS and BFS.
ii) Explain the Minimax Algorithm with example
iii) Explain Genetic Algorithm with example
iv) Explain Ao* algorithm with example.
v) Explain the concept of problem decomposition in AI.

3. Attempt **any three** of the following. **12**

i) Explain Backward Chaining with example
ii) What is Rete Net? Explain with neat diagram
iii) What is Goal Stack Planning and Explain its applications in AI.
iv) Explain the Ant Colony Optimization algorithm

4. Attempt **any two** of the following. **12**

i) Explain the Turing Test with neat diagram
ii) Explain Beam Search algorithm.
iii) Explain Hill Climbing algorithm

5. Attempt **any two** of the following. 12

(i) Explain Forward chaining algorithm with example
(ii) Explain the alpha-beta pruning algorithm.
iii) What is AI? Explain any 3 characteristics of AI.

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